

Perfect 🚀

Here's your **2-month (8-week) day-wise ChatGPT Maths for ML/AI/DL learning plan**.
Each day you'll **copy-paste prompts** into ChatGPT, and I'll act as your teacher/tutor/coding partner.



2-Month ChatGPT Maths Plan for ML/AI/DL

Week 1 – Foundations

Day 1: Algebra basics

👉 Prompt: *"Teach me variables, equations, and functions with ML examples."*

👉 Practice prompt: *"Give me 10 problems on solving linear equations with solutions."* ✓

Day 2: Exponents & logarithms

👉 *"Explain exponentials & logs with examples in ML (learning rates, softmax)."* ✓

Day 3: Functions & transformations

👉 *"Explain linear, quadratic, exponential functions with ML applications."* ✓

Day 4–5: Coordinate geometry

👉 *"Teach me slope, distance, dot product using vector geometry."* ✓

Day 6–7: Revision + problem set

👉 *"Quiz me on algebra + geometry with 15 mixed problems."* ✗ not completed

Week 2–3 – Linear Algebra

Day 8–9: Vectors

👉 *"Teach me dot product, norms, cosine similarity with ML applications."* ✓

Day 10–11: Matrices

👉 *"Explain matrix multiplication with neural network examples."* ✓

Day 12–13: Special matrices (identity, diagonal, orthogonal)

👉 *"Show me why identity matrix is important in ML."*

Day 14–16: Eigenvalues & eigenvectors

👉 *“Explain eigenvalues/eigenvectors with PCA example.”*

Day 17–18: Matrix decompositions (SVD, LU)

👉 *“Teach me SVD with recommender system example.”*

Day 19–20: Coding practice

👉 *“Write Python code to implement PCA from scratch.”*

Week 4–5 – Calculus

Day 21–22: Limits & derivatives

👉 *“Explain derivatives visually + give 10 practice problems.”*

Day 23–24: Partial derivatives

👉 *“Show me how partial derivatives are used in gradient descent.”*

Day 25–26: Gradients & Jacobians

👉 *“Teach me gradient + Jacobian with ML applications.”*

Day 27–28: Chain rule & backpropagation

👉 *“Walk me through backprop step-by-step with an example NN.”*

Day 29–30: Integration basics

👉 *“Explain integration for probability distributions (e.g. Gaussian).”*

Week 6 – Probability & Statistics

Day 31–32: Probability basics

👉 *“Explain conditional probability & Bayes’ theorem with spam filter example.”*

Day 33–34: Random variables + distributions

👉 *“Teach me Gaussian, Bernoulli, Binomial, Poisson with ML use cases.”*

Day 35–36: Expectation, variance, covariance

👉 *“Explain covariance & correlation in ML with dataset examples.”*

Day 37–38: MLE (Maximum Likelihood Estimation)

👉 *“Teach me MLE with logistic regression example.”*

Day 39–40: Hypothesis testing

👉 *“Explain hypothesis testing with A/B testing example.”*

Week 7 – Optimization

Day 41–42: Convex functions & optimization

👉 *“Explain convex optimization with gradient descent example.”*

Day 43–44: Gradient descent + variants

👉 *“Teach me SGD, Momentum, RMSProp, Adam with intuition.”*

Day 45–46: Constrained optimization

👉 *“Explain Lagrange multipliers with ML example.”*

Day 47–48: Coding practice

👉 *“Write Python code for gradient descent on linear regression.”*

Week 8 – Advanced + Applications

Day 49–50: Information theory

👉 *“Explain entropy & KL divergence with ML examples.”*

Day 51–52: Cross-entropy loss

👉 *“Show me cross-entropy loss in classification with Python code.”*

Day 53–54: Numerical methods

👉 *“Explain floating point errors in ML + show Python example.”*

Day 55–56: Graph theory basics

👉 *“Teach me graphs & adjacency matrices with GNN example.”*

Day 57–58: Full ML application

👉 *“Walk me through maths behind neural networks step by step.”*

Day 59–60: Revision & Final Challenge

👉 *“Quiz me with 30 mixed questions across algebra, linear algebra, calculus, probability, optimization.”*

👉 *“Help me implement a mini neural net from scratch with only numpy.”*



How to Follow

- Every day: Copy the **main prompt** + **practice prompt**.
- Do exercises → get feedback → code → connect to ML.
- Keep notes in one place (Notion, Obsidian, Google Docs).

Would you like me to also create a **single reusable “prompt template sheet”** (like flashcards) that you can use for *any topic*, so you don’t have to think of what to ask each time?